Module 1: Basic Physiology

TOPIC 1: CELL PHYSIOLOGY

>. Organelle Function (Apoptosis)

- **Question:** Which organelle plays a central role in the intrinsic pathway of apoptosis? (Midterm 2025)
- **Answer: B) Mitochondrion**
- **Guyton Explanation:** The intrinsic pathway of apoptosis is initiated when **Mitochondria** release cytochrome c into the cytoplasm, which activates caspases.

>. Organelle Function (Protein Synthesis)

- **Question:** Which of the following is the function of rough endoplasmic reticulum? (Mock 2024, Mock 2023)
- **Answer: b) Synthesis of proteins**
- **Guyton Explanation:** The RER is studded with ribosomes and is the site where new protein molecules are synthesized and extruded into the endoplasmic matrix.

>. Organelle Function (Self-Replication)

- **Question:** Which of the following structure regenerates and replicates spontaneously? (Mock 2023)
- **Answer: a) Mitochondria**
- **Guyton Explanation:** Mitochondria have their own DNA and can replicate themselves (self-replicate) to meet the cell's energy needs.

>. Organelle Content (Catalase)

- **Question:** Which of the following organelles contains catalase? (Mock 2019)
- **Answer: a) Peroxisomes**
- **Guyton Explanation:** Peroxisomes contain oxidases and **catalase**, which break down hydrogen peroxide and other toxins.

>. Cell Structure (Cilia)

- **Question:** Which of the following are most abundantly found in cilia of a cell? (Final 2021)
- **Answer: d) Microtubules**
- **Guyton Explanation:** The core of a cilium (the axoneme) is composed of an array of **microtubules** (9 doublets + 2 central singles) which allow it to move.

>. Cell Membrane Structure

- **Question:** Which component of the cell membrane primarily acts as a selective barrier to polar substances? (Midterm 2025)
- **Answer: B) Phospholipid bilayer**
- **Guyton Explanation:** The lipid bilayer is impermeable to water-soluble (polar) substances like ions and glucose, acting as the primary barrier.

TOPIC 2: TRANSPORT ACROSS MEMBRANES

. Active Transport Mechanism

- **Question:** Sodium-Potassium pump is an example of: (Midterm 2020)
- **Answer: b) Primary active transport**

- **Guyton Explanation:** The Na+-K+ pump uses energy *directly* from ATP breakdown to pump ions against their gradients.

. Na+-K+ Pump Stoichiometry

- **Question:** Regarding Na-K ATPase which of the following is a correct statement: (Mock 2024)
- **Answer: e) It pumps out 3 Na-ions and takes in 2 K-ions.**
- **Guyton Explanation:** The pump moves **3 Sodium ions OUT** and **2 Potassium ions IN** for every ATP hydrolyzed.

. Facilitated Diffusion Requirements

- **Question:** Facilitated diffusion requires: (Mock 2019, Midterm 2020)
- **Answer: c) Carrier**
- **Guyton Explanation:** Facilitated diffusion (carrier-mediated diffusion) requires a specific **carrier protein** to transport the substance but does *not* use ATP (passive).

. Glucose Transport

- **Question:** By which mechanism is glucose absorbed from the lumen of the small intestine into enterocytes? (Midterm 2025, Midterm 2022)
- **Answer: D) Secondary active transport (Co-transport)**
- **Guyton Explanation:** Glucose enters the cell against its gradient by riding along with Sodium (Sodium-Glucose Co-transport), which is powered by the Sodium gradient established by the Na-K pump.

. Endocytosis (Bacteria)

- **Question:** The form of endocytosis in which whole cell such as bacteria are brought in is: (Mock 2019)
- **Answer: d) Phagocytosis**
- **Guyton Explanation:** Phagocytosis is the ingestion of large particles (like bacteria), whereas pinocytosis is the ingestion of fluid/small molecules.

. Diffusion Properties

- **Question:** Substance easily diffusable across plasma membrane? (Midterm 2022)
- **Answer: b) Fat soluble**
- **Guyton Explanation:** Lipid-soluble substances (like O2, CO2, alcohol) diffuse directly through the lipid bilayer with ease.

TOPIC 3: BODY FLUIDS & HOMEOSTASIS

>. Fluid Compartments (Volume)

- **Question:** Which body fluid compartment contains the largest volume of water? (Midterm 2025)
- **Answer: C) Intracellular fluid**
- **Guyton Explanation:** Intracellular fluid makes up about 40% of body weight (approx. 28 Liters in a 70kg man), whereas ECF is only 20% (14 Liters).

>. Fluid Compartments (Composition)

- **Question:** As compared to extracellular fluid, intracellular fluid contains: (Mock 2024)
- **Answer: a) Higher concentration of potassium ions.**
- **Guyton Explanation:** ICF is rich in **Potassium**, Magnesium, and Phosphate. ECF is rich in Sodium, Chloride, and Bicarbonate.

>. Fluid Composition Differences

- **Question:** Plasma fluid differ from interstitial fluid in concentration of which of the following? (Mock 2024, Mock 2023)

- **Answer: e) Proteins**
- **Guyton Explanation:** Plasma and Interstitial fluid have very similar ionic compositions, but **Plasma has a much higher protein concentration** because capillaries are not permeable to proteins.

>. Positive Feedback

- **Question:** Which of the following is an example of a positive feedback mechanism? (Mock 2024, Final 2021, Midterm 2022)
- **Answer: Child birth (Labor contractions)**
- **Guyton Explanation:** Most body systems use *Negative Feedback* (correction). *Positive Feedback* is a vicious cycle (amplification), seen in Blood Clotting, Childbirth (Oxytocin), and the upstroke of an Action Potential.

>. Homeostasis Definition

- **Question:** Which of the following is best description of Homeostasis? (Mock 2024)
- **Answer: c) Body systems working co-operatively to maintain body function within normal limits** (Maintenance of nearly constant conditions in the internal environment).

>. Body Fluid Calculation

- **Question:** The body fluid of 42 litres in a young man of 70 kg body weight has following fluid distribution... (Final 2021)
- **Answer: e) Intracellular fluid is 28 liter**
- **Guyton Explanation:** Total Body Water (42L) = ICF (2/3 = 28L) + ECF (1/3 = 14L). ECF = Interstitial (11L) + Plasma (3L).

TOPIC 4: BUFFERS (General Phys)

>. Interstitial Buffer

- **Question:** Which of the following is the principal buffer in interstitial fluid? (Mock 2024, Mock 2023)
- **Answer: c) Bicarbonate** (Carbonic acid/Bicarbonate system)
- **Guyton Explanation:** The Bicarbonate buffer system is the most important quantitative buffer in the **Extracellular Fluid** (including Interstitial fluid). Proteins are the major buffer in cells and plasma, but less so in interstitial fluid due to low concentration.

>. Edema Source

- **Question:** Proteins in edema fluid is provided by which of the following? (Mock 2024, Mock 2023)
- **Answer: a) Blood** (Plasma)
- **Guyton Explanation:** Edema fluid is essentially plasma that has filtered out of capillaries. If there is high protein in edema (exudate), it leaked from the **Blood** due to increased capillary permeability.